

Technology Stack

Date	15 March 2026
Team ID	SWTID-2026-2085
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side <i>(e.g., React, Angular, HTML/CSS)</i>	HTML5, CSS3, Bootstrap, JavaScript	Provides a responsive, user-friendly interface for farmers to enter soil and environmental data and view crop recommendations.

architecture and provide a brief justification for your choice.



2

SMART

**Backend /
Server-Side**
Let's Bridge the Gap

Python, Flask

Flask is lightweight and integrates easily with Machine Learning models, enabling fast and efficient prediction processing.

**SKILL
WALLET**
A SmartBridge Product

*(e.g., Node.js,
Python/Django, Java)*

3

**Database / Data
Storage**CSV Dataset
(Crop
Recommendation
Dataset)

Stores crop training data used by the Machine Learning model. It is simple, lightweight, and sufficient for the project.

*(e.g., MySQL,
MongoDB,
PostgreSQL)*

4

**Cloud / Hosting /
Deployment**CSV Dataset
(Crop
Recommendation
Dataset)

Enables easy deployment and online access to the application with minimal configuration.
*(Use the one you actually
deployed.)*

*(e.g., AWS, Azure,
Heroku, Docker)*

5

**Version Control &
CI/CD***Git, GitHub]*

Tracks project versions, enables collaboration, and securely stores the project source code.

6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	Scikit-learn, Pandas, NumPy, Matplotlib, Pickle	Used for data preprocessing, Machine Learning model training, prediction generation, visualization, and model serialization.
----------	---	---	--

Technology Stack Details

